

Vehicle Currencies in AMMs: Model and Empirical Review Packet

Purpose: internal review before manuscript drafting.

Current status: no additional broad experiments are needed before writing if the paper is written around bounded claims. The remaining task is table hierarchy and claim discipline.

Bounded empirical spine:

1. Measurement: vehicle use is conditional on indirect routing and must be paired with all-route vehicle share.
2. P1/setup: direct-market incompleteness is descriptive scope for the central propositions.
3. P2: vehicle intermediation and vehicle-linked LP liquidity are mutually persistent.
4. P3: credibility or risk shocks to a vehicle reduce its route advantage and rotate order flow toward substitutes.
5. P4a: market-architecture changes that deepen direct pairwise markets reduce reliance on vehicle routes.
6. P4b: settlement netting raises LP willingness to supply vehicle-linked liquidity where routed demand is present.

Implementation discipline: WETH, stablecoins, concentrated-liquidity launch, and flash-accounting launch are empirical test beds.

They should appear in table labels and identification text, not as the propositions themselves.

Model Structure

Trader chooses between direct route $i \rightarrow j$ and vehicle route $i \rightarrow k \rightarrow j$.

$$\text{DirectCost}(q) = f_D + s_D + \theta q / LD$$

$$\text{VehicleCost}(q) = f_{IK} + f_{KJ} + s_K + \rho_K + \theta q (1/L_{IK} + 1/L_{KJ})$$

$$\text{RouteAdvantage} = \text{DirectCost} - \text{VehicleCost}$$

$$\text{BridgeShare}(\delta) = 1 / (1 + \exp(-\lambda \delta))$$

P1 is treated as setup: direct-market weakness explains why vehicle routes are relevant; direct-unavailable cases are partly mechanical.

P2 is reduced-form liquidity-route feedback:

$$E[\text{BridgeShare}_{\{k,t+h\}}] = \alpha_k + \beta_L * \text{LPConcentration}_{\{k,t\}} + \rho * \text{BridgeShare}_{\{k,t\}}.$$

$$E[\text{LPConcentration}_{\{k,t+h\}}] = \alpha_l + \beta_B * \text{BridgeShare}_{\{k,t\}} + \psi * \text{LPConcentration}_{\{k,t\}}.$$

P3 is interval-neutral in theory: vehicle risk or credibility cost lowers route advantage on impact. Same-day stress episodes are the empirical implementation.

P4a: higher direct-route liquidity reduces relative vehicle-route reliance.

P4b: netting lowers the operational inventory cost of supporting routed demand, increasing optimal vehicle-linked LP supply.

Mathematica Derivation Highlights

Key formal signs from output/model/model_derivations.txt:

P1AvailabilityProtection:

- $d\text{BridgeShare}/d\text{Advantage} = \lambda * \text{Sech}[\dots]^2 / 4 > 0$.
- VehicleUsefulCondition includes direct-unavailable, thin-direct, and positive-advantage cases.

P2LiquidityRouteFeedback:

- $d\text{Advantage}/d\text{LIK} > 0$ and $d\text{Advantage}/d\text{LKJ} > 0$.
- $d\text{ExpectedBridgeShareNext}/d\text{VehicleLiquidity} = \beta_L \geq 0$.
- $d\text{ExpectedBridgeShareNext}/d\text{CurrentBridgeShare} = \rho \geq 0$.
- $d\text{ExpectedVehicleLiquidityNext}/d\text{CurrentBridgeShare} = \beta_B \geq 0$.

P3StressRotation:

- $d\text{Advantage}/d\text{VehicleRisk} = -1$.
- $d\text{BridgeShare}/d\text{VehicleRisk} < 0$.

P4aDirectMarketDeepening:

- $d\text{Advantage}/d\text{DirectLiquidityMultiplier} < 0$.

P4bSettlementNettingLiquidity:

- $\text{PhysicalVehicleMovement} = (1 - n) * \text{GrossVehicleExposure}$.
- $\text{LPPayoff} = \text{fee value from routed demand} - \text{netting-sensitive operating cost} - \text{convex liquidity cost}$.
- $d\text{OptimalVehicleLiquidity}/d\text{Netting} = \kappa_1 / \chi \geq 0$.
- $d\text{OptimalVehicleLiquidity}/d\text{RoutedDemand} = \phi / \chi > 0$.

Full derivation object:

Model derivation highlights

P2 liquidity-route feedback:

$\text{ExpectedBridgeShareNext} = \alpha_K + \beta_L * \text{VehicleLiquidity} + \rho * \text{CurrentBridgeShare}$.

$d\text{ExpectedBridgeShareNext}/d\text{VehicleLiquidity} = \beta_L \geq 0$.

$d\text{ExpectedBridgeShareNext}/d\text{CurrentBridgeShare} = \rho \geq 0$.

$\text{ExpectedVehicleLiquidityNext} = \alpha_L + \beta_B * \text{CurrentBridgeShare} + \psi * \text{VehicleLiquidity}$.

$d\text{ExpectedVehicleLiquidityNext}/d\text{CurrentBridgeShare} = \beta_B \geq 0$.

$d\text{ExpectedVehicleLiquidityNext}/d\text{VehicleLiquidity} = \psi \geq 0$.

This is the model object for a Matthew-effect interpretation: liquidity predicts future route use, and route use predicts future liquidity.

P4b settlement netting and LP supply:

$\text{PhysicalVehicleMovement} = (1 - n) * \text{GrossVehicleExposure}$.

$\text{LPPayoff} = \phi * \text{RoutedDemand} * \text{VehicleLiquidity} - (\kappa_0 + \kappa_1 * (1 - n)) * \text{VehicleLiquidity} - \chi * \text{VehicleLiquidity}^2 / 2$.

$\text{OptimalVehicleLiquidity} = (\phi * \text{RoutedDemand} - \kappa_0 - \kappa_1 * (1 - n)) / \chi$, truncated at zero.

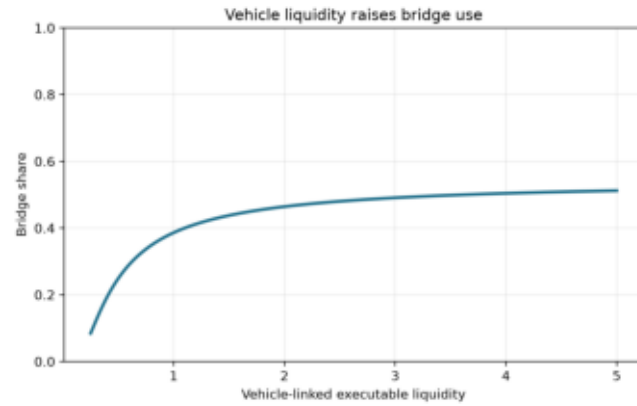
$d\text{OptimalVehicleLiquidity}/d\text{Netting} = \kappa_1 / \chi \geq 0$.

$d\text{OptimalVehicleLiquidity}/d\text{RoutedDemand} = \phi / \chi > 0$.

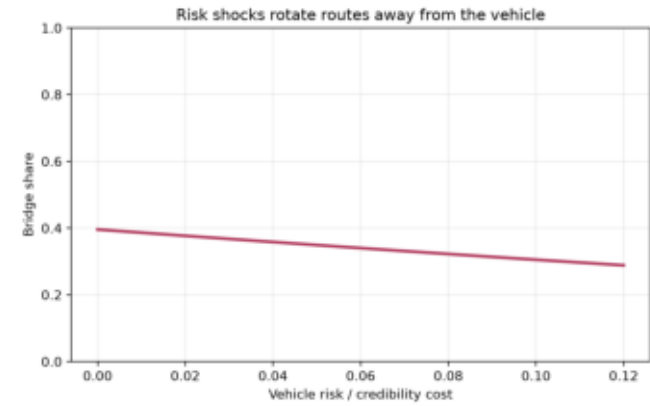
The behavioral prediction goes beyond the accounting identity that netting lowers transfers: netting raises LP willingness to supply vehicle-linked liquidity where routed demand is present.

Numerical Simulations

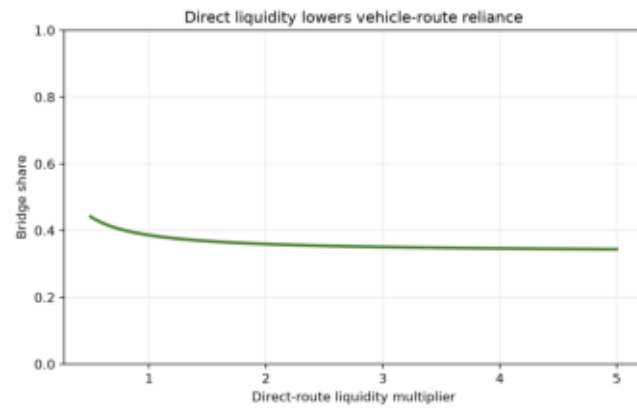
bridge_share_liquidity



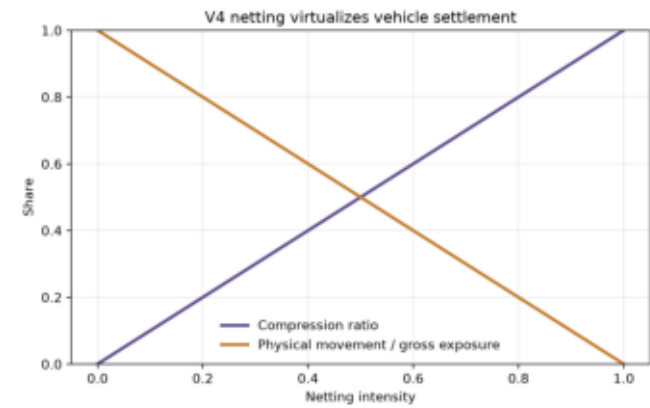
bridge_share_risk



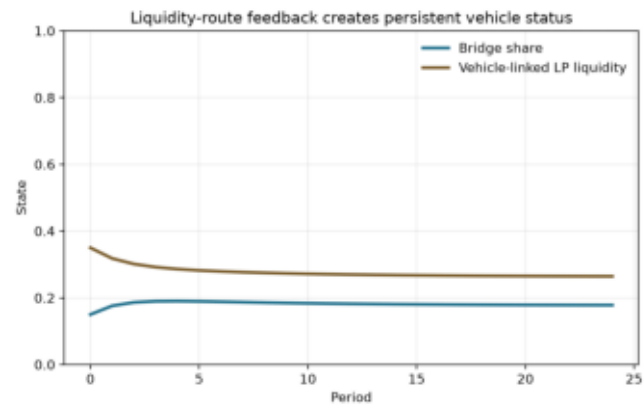
bridge_share_direct_liquidity



v4_netting_compression



liquidity_route_feedback



netting_lp_supply

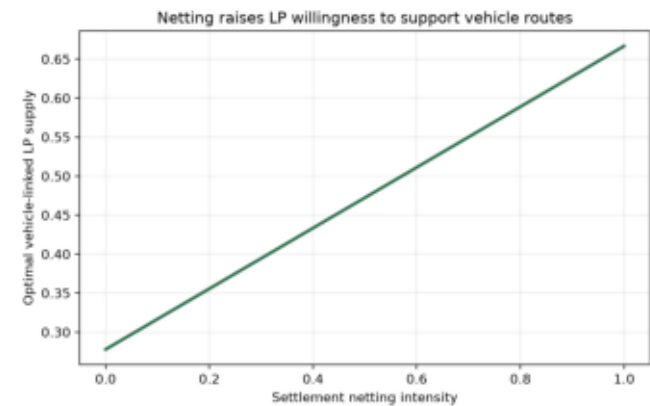


Table 1. Measurement and Scope

Panel		Quantity	Main estimate	Scope / companion estimate
A. Vehicle-use denominators, 2026		WETH	44.5% conditional BridgeShare	5.3% all-route bridge share
50.0% endpoint-pair coverage				
A. Vehicle-use denominators, 2026		USDC	20.6% conditional BridgeShare	2.5% all-route bridge share
25.3% endpoint-pair coverage				
A. Vehicle-use denominators, 2026		USDT	23.6% conditional BridgeShare	3.0% all-route bridge share
20.7% endpoint-pair coverage				
B. Exact-quote scope	Exact-quote covered share		78.9	NaN Unified leg-volume share covered by V2/Sushi V2/V3 plus Bal
B. Exact-quote scope	Excluded Curve+Fluid share		16.2	NaN Unified leg-volume share not covered by
B. Exact-quote scope	Excluded / covered ratio		0.205	NaN Maximum exact-quote scope exposure relativ
B. Exact-quote scope	Excluded stable-leg share		75.3	NaN Excluded venues are stablecoin-heavy, so ro

Table 2. P1 Route Availability and Thin-Direct Protection

Trade size	Direct available (%)	WETH route available (%)	No-direct / WETH-available rows	Thin-direct median advantage (bp)	
High-quality-direct	median advantage (bp)	Common-support	median (bp)		
\$1,000	72.1	67.4	9,584	142.65	8.95
\$10,000	72.1	67.4	9,584	190.21	
-1.12	21.52				
\$100,000	72.1	67.4	9,584	349.28	
-76.16	-0.39				

Table 3. P2 Liquidity-Route Feedback

Panel	Horizon	Outcome	Main regressor	Beta	SE	t	p	Control
A. Liquidity -> future intermediation BridgeShare beta 0.460	t+7	future BridgeShare	LP concentration	0.142	0.013	10.57	<0.001	current
B. Intermediation -> future liquidity concentration beta 0.841	t+7	future LP concentration	current BridgeShare	0.053	0.010	5.23	<0.001	LP
B. Intermediation -> future liquidity concentration beta 2.717	t+7	future log LP liquidity	current BridgeShare	1.696	0.067	25.29	<0.001	LP
A. Liquidity -> future intermediation BridgeShare beta 0.330	t+30	future BridgeShare	LP concentration	0.148	0.013	11.11	<0.001	current
B. Intermediation -> future liquidity concentration beta 0.744	t+30	future LP concentration	current BridgeShare	0.078	0.013	6.12	<0.001	LP
B. Intermediation -> future liquidity concentration beta 2.401	t+30	future log LP liquidity	current BridgeShare	1.760	0.074	23.87	<0.001	LP

Table 4. P3 Vehicle Risk and Stress Rotation

Panel	Estimate	Events	Effect	t	p	Interpretation
A. Main decomposed same-day effect prior-28-day baseline	WETH share change	20	-1.48 pp	-2.59	0.018	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	Stable share change	20	1.48 pp	2.59	0.018	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	WETH-minus-stable change	20	-2.96 pp	-2.59	0.018	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	Aggregate direct-route share change	20	-0.25 pp	-0.42	0.679	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	Log indirect-route volume change	20	0.51 log points	3.20	0.005	event-day minus
B. Threshold/overlap sensitivity negative share 67.0%	all events, threshold 6%	91	-3.12 pp	-3.44	<0.001	
B. Threshold/overlap sensitivity negative share 65.3%	all events, threshold 8%	49	-2.95 pp	-3.41	0.001	
B. Threshold/overlap sensitivity negative share 65.4%	all events, threshold 10%	26	-3.74 pp	-3.70	0.001	
B. Threshold/overlap sensitivity negative share 63.2%	all events, threshold 12%	19	-3.18 pp	-2.68	0.015	
B. Threshold/overlap sensitivity negative share 61.1%	top 20, threshold 8%	18	-3.09 pp	-2.48	0.024	
B. Threshold/overlap sensitivity negative share 57.6%	non-overlap, threshold 8%	33	-3.01 pp	-2.45	0.020	

Table 5. P4a Direct-Market Deepening and Vehicle-Route Opportunity

	Outcome	Rows	Pairs	Post-V3	effect	t	p	Pretrend slope	Pretrend t	Pretrend p	Units
No-direct WETH	availability	1,822	190		-25.81	-7.62	<0.001	0.071	0.1	0.922	pp

Table 6. P4b Settlement Netting and LP Response

	Panel	Sample / diagnostic	Cells	V3	V4	Difference / balance	p
A. Transfer incidence by route-size bin		All	500	100.0%	81.4%	-18.6	
pp							
A. Transfer incidence by route-size bin		Route size: Small	104	100.0%	60.6%	-39.4	
pp							
A. Transfer incidence by route-size bin		Route size: Medium	73	100.0%	87.7%	-12.3	
pp							
A. Transfer incidence by route-size bin		Route size: Large	94	100.0%	96.8%	-3.2	
pp							
	B. Matched-sample balance	uniswap_v3	500	\$6,136	NaN	53.05	100.0
	B. Matched-sample balance	uniswap_v4	500	NaN	\$3,542	71.05	81.4
	B. Matched-sample balance	V4 - V3 within cell	500	NaN	NaN	log route diff=-0.692	t=-7.58, p=<0.001
C. LP response by netting exposure		LP concentration	630	NaN	NaN	post x netting exposure beta -0.063 (t=-10.12)	<0.001
C. LP response by netting exposure		log LP liquidity	630	NaN	NaN	post x netting exposure beta 2.132 (t=9.47)	<0.001

Table 7. Specification Registry

Test	Unit	Sample	Outcome	Reg
P1 availability/thin-direct endpoint-pair x day x trade size	V2/Sushi V2/V3 exact quoteable venues			
route availability and WETH advantage	WETH vehicle route	endpoint-pair-day aggregation		
direct available 72.1% 9,584 no-direct rows; thin-direct 142.65-349.28 bp		availability and thin-direct		
execution protection				
P2 liquidity-route feedback	token x day	vehicle candidates future BridgeShare;		
future LP concentration/liquidity LP concentration; current BridgeShare	token/date FE; date-clustered SE			
see Table m03	two-way coefficients positive; p<0.001	reduced-form liquidity-route		
feedback				
P3 impact stress event x common endpoint-pair set	WETH downside event days	WETH-minus-stable BridgeShare	event-day W	
P4a V3 opportunity	endpoint-pair x month	balanced V3 launch-window pairs	no-direct/WETH-available indicator	
P4b V4 settlement matched route unit; token x week matched V3/V4 cells; LP panel around launch		transfer incidence; LP liquidity response V4	route unit; post	